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Ensuring Safe Hydrogen Storage: Advanced Numerical Modelling of Caprock Integrity and Reservoir Geomechanics for Hydrogen Containment and Leakage Prevention

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Abstract

Underground hydrogen storage (UHS) is a key technology for large-scale, long-duration energy storage, supporting seasonal balancing of renewable generation and demand. Geological options such as saline aquifers, depleted reservoirs, and salt caverns offer significant capacity, yet cyclic injection and withdrawal can trigger complex hydro-mechanical responses that affect both performance and containment integrity. In this study, a fully coupled two-dimensional poromechanical model was developed to evaluate the stability of a homogeneous saline aquifer under repeated hydrogen cycling. The framework integrates a pressure equation, well representation, and Biot's poroelastic formulation with an explicit creep deformation component, enabling the simulation of pore pressure, stresses, volumetric strain, and property evolution. Flow simulations showed strongly localized pressure build-up and smooth radial dissipation, mirrored by the volumetric strain field. Laboratory observations further confirmed the temperature sensitivity of creep and the role of mineral heterogeneity in accelerating fracture nucleation. Single-cycle simulations revealed localized property changes near the injector, with permeability enhancement driven by elastic dilation, while two-cycle analyses demonstrated cumulative effects, including residual porosity and up to 27% permeability increase at the injector. Incorporating creep highlighted a ratcheting mechanism, whereby volumetric strain and permeability changes accumulated across cycles even under constant loading magnitudes. These results demonstrate the necessity of accounting for both time-dependent deformation and flow-mechanical coupling when evaluating UHS performance, as cyclic operations can induce progressive, spatially localized changes that influence injectivity and long-term containment.

Introduction

Underground hydrogen storage (UHS) is increasingly recognized as a cornerstone of future low-carbon energy systems, offering the capacity to balance intermittent renewable generation with large-scale, long-duration demand (Morales et al., 2024). Geological structures such as saline aquifers, depleted hydrocarbon reservoirs, and salt caverns provide attractive options for storage, yet their utilization requires careful evaluation of geomechanical stability and long-term containment security (Trushko and Protosenya, 2019;

Aljaberi et al., 2025). Cyclic injection and withdrawal operations subject formations to fluctuating pore pressures and stress regimes, driving coupled hydro-mechanical responses that control injectivity, recovery efficiency, and system integrity (Houhou and Laloui, 2022).

Most existing studies have focused on elastic deformation, investigating stress redistribution and matrix response under operational cycling (Rodriguez et al., 2024; Rodriguez et al., 2022). However, the influence of inelastic mechanisms, particularly time-dependent creep deformation, remains insufficiently addressed in the hydrogen storage context. Creep can progressively modify stress distribution, porosity, and permeability, thereby altering containment performance and long-term mechanical stability (Ciancimino et al., 2024; Hameli et al., 2024). These effects become especially relevant under sustained pressurization or frequent cycling, conditions in which viscoelastic and viscoplastic mechanisms may be activated in the rock mass. Neglecting such time-dependent processes risks underestimating the extent of irreversible deformation and the potential for long-term property evolution in storage formations (Anemangely, Ramezanzadeh and Behboud, 2019).

In cyclic hydrogen storage operations, creep deformation interacts with both injection-induced loading and withdrawal-induced unloading phases, producing cumulative effects on sealing capacity and structural stability. Mechanisms such as gradual closure or dilation of microfractures, slow shear displacement along bedding planes, and progressive compaction or expansion of the caprock matrix represent creep-driven processes that can modify containment performance over the operational lifespan of a storage facility. To address this challenge, the present study incorporates a geomechanical modeling framework that explicitly accounts for both elastic and inelastic deformation, thereby providing a more comprehensive assessment of long-term storage integrity under realistic operational conditions.

Recent investigations by Kumar and co-workers (Kumar et al., 2020; Kumar et al., 2021; Kumar et al., 2022) have advanced the understanding of creep behavior in sandstone, shale, and salt formations relevant to hydrogen storage. Their work introduced a series of constitutive models, validated through laboratory experiments and numerical simulations, capable of reproducing creep responses under storage-representative stress and temperature regimes. These models were subsequently applied to field-scale scenarios, offering valuable insights into the role of time-dependent deformation in geological storage systems. The numerical framework proposed by Kumar et al. (2021) has since provided a foundation for further modeling developments, particularly with adaptations designed for sandstone formations.

Mineralogical composition has been demonstrated to exert significant control on creep behavior in porous media. Hansen et al. (1987) reported that small proportions of clay and anhydrite in fine-grained salt had minimal influence on steady-state creep rates. In contrast, Munson et al. (1999) observed that even trace levels of impurities could reduce creep deformation relative to pure halite. Subsequent work by Hunsche et al. (1996, 1998) concluded that impurities act as barriers to dislocation motion, requiring higher applied stresses to sustain comparable creep rates. Extending this perspective to sandstones, Shuai et al. (2023) applied microscopic imaging techniques to analyze crack propagation and demonstrated that mineralogical heterogeneity and grain-scale variability promote fracture nucleation and accelerate the transition from microcracking to macroscopic failure. A recent contribution by AlDhuhoori et al. (2024) further examined the influence of mineralogical composition on the geomechanical fabric of rocks relevant to hydrogen storage. This study emphasized the geomechanical properties of constituent minerals and their implications for site selection. Using creep deformation simulations of sandstone samples with varied mineral compositions, the authors quantified strain evolution under cyclic loading. Homogeneous calcite-rich samples displayed the greatest resistance to creep deformation after three loading cycles, attributed to calcite's high Young's modulus. For heterogeneous cases, a calcite-quartz mixture exhibited the highest resistance to time-dependent deformation. Among different lithologies, basalt was identified as the most creep-resistant, while siltstone proved the most susceptible. The findings further revealed that rock stability declines with increasing cycle number, underscoring the importance of optimizing operational cycling to preserve long-term containment.

To address this gap, the present study adopts a multi-faceted approach that combines laboratory-scale experimental analysis and numerical modeling of a field-scale study. At the laboratory scale, controlled experiments provide insights into pore-scale deformation, permeability evolution, and creep behavior under hydrogen storage conditions, offering direct validation for numerical predictions (AIDhuhoori et al., 2020; AIDhuhoori et al., 2023a). Numerically, a coupled flow-geomechanics model is developed to capture both the elastic and inelastic components of rock deformation, enabling quantitative assessment of pore pressure response, volumetric strain, stress redistribution, and permeability changes during injection-withdrawal cycles (Nauroy, 2011). At the field scale, this modeling framework is extended to evaluate the performance of an aquifer storage system under operationally realistic boundary conditions, providing an integrated perspective on containment efficiency and long-term stability. By bridging laboratory experiments, numerical modeling, and field-scale application, the study offers a comprehensive evaluation of the coupled processes governing underground hydrogen storage. The findings highlight the interplay between instantaneous elastic responses and time-dependent creep effects, emphasizing the need to account for inelastic deformation in both design and operational planning of large-scale hydrogen storage projects.

Methodology

The mechanical response of subsurface formations during injection and withdrawal operations is governed by the coupled interaction between fluid flow and rock deformation. In this study, a two-dimensional, fully coupled poromechanical model was developed to evaluate the geomechanical stability of an aquifer subject to cyclic hydrogen injection and production. The formulation integrates governing flow equations, pressure equation for fluid flow, relative permeabilities, well representation using source-sink terms, and a mechanical equilibrium model incorporating time-dependent creep deformation. With multiphase effects and saturation changes being included, the essential hydro-mechanical coupling mechanisms are preserved, enabling a robust analysis of stress redistribution, deformation, and the evolution of reservoir properties and fluid flow in the porous media.

Pressure Equation

The pressure is governed by the mass conservation equation for a slightly compressible, single-phase fluid in a porous medium, coupled with Darcy's law. For constant total compressibility, the governing equation in two dimensions is expressed as:

$$\phi c_t \frac{\partial P}{\partial t} = \nabla \cdot (k \lambda_t \nabla P) + q \quad (1)$$

where ϕ is porosity, c_t is the total compressibility, k is the absolute permeability tensor, λ_t is total mobility, P is pore pressure, and q is the volumetric source/sink term.

The equation is discretized using the finite volume method, and time integration is performed using an implicit scheme to enhance stability, particularly in stiff systems where mechanical feedback is significant. The computed pressure from Equation (1) is directly coupled to the mechanical equilibrium equations via Biot's coefficient.

Governing Equations

The simulator solves the two-phase flow equations in porous media:

$$\frac{\partial (\phi \rho_i S_i)}{\partial t} + \nabla \cdot (\rho_i v_i) = q_i \quad (2)$$

With Darcy's flow equation being represented in the following equation:

$$v_i = -k \lambda_i \nabla (p - \rho_i g z) \quad (3)$$

And the saturation equation is presented in the following equation:

$$\phi \frac{\partial S_w}{\partial t} = -\nabla \cdot (f_w v_t) + q_w - \phi S_w (c_w + c_\phi) \frac{\partial p}{\partial t} \quad (4)$$

Relative Permeability

To represent the relative permeability changes, Corey's model has been used:

$$K_{rw} = S_w^{*n_w} \quad (5)$$

$$K_{rg} = (1 - S_w^*)^{n_g} \quad (6)$$

Well Representation

Well operations are incorporated into Equation (1) through the source/sink term q , defined according to operational control parameters: either by a fixed rate or a prescribed bottomhole pressure (BHP). For rate-controlled wells:

$$q = q_{spec} \quad (7)$$

where q_{spec} is the specified volumetric injection or production rate.

For pressure-controlled wells, the flow rate is calculated using Peaceman's formulation:

$$q = WI \cdot \lambda_t \cdot (P_{cell} - P_{bhp}) \quad (8)$$

Here, WI is the well index, P_{cell} is the pressure at the well block, and P_{bhp} is the prescribed BHP.

The well index in two dimensions is computed as:

$$WI = \frac{2\pi K_h h}{\ln\left(\frac{r_e}{r_w}\right)} \quad (9)$$

where K_h is the effective horizontal permeability, h is formation thickness, r_e is the drainage radius, and r_w is the wellbore radius. The resulting modifies the pressure distribution, which in turn influences the mechanical response through poroelastic coupling.

Geomechanical Formulation and Creep Deformation

The mechanical equilibrium is described by Biot's theory of linear poroelasticity, assuming quasi-static deformation and small strains. The equilibrium condition is expressed as:

$$\nabla \cdot \sigma + f = 0 \quad (10)$$

where f is the body force arising from pore pressure gradients:

$$f = -\alpha \nabla P \quad (11)$$

Here, α is the Biot coefficient. The total stress tensor is given by:

$$\sigma_{ij} = \lambda \varepsilon_v \delta_{ij} + 2G \varepsilon_{ij} - \alpha P \delta_{ij} \quad (12)$$

with volumetric strain $\varepsilon_v = \varepsilon_{xx} + \varepsilon_{yy}$. The effective stress is defined as:

$$\sigma'_{ij} = \sigma_{ij} + \alpha P \delta_{ij} \quad (13)$$

Strains follow the small-strain assumption:

$$\varepsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \quad (14)$$

The coupled Navier-Cauchy equations in two dimensions are:

$$(\lambda + 2G)\frac{\partial^2 u_x}{\partial x^2} + G\frac{\partial^2 u_x}{\partial y^2} + (\lambda + 2G)\frac{\partial^2 u_y}{\partial x \partial y} + \alpha \frac{\partial P}{\partial x} = 0 \quad (15)$$

$$(\lambda + 2G)\frac{\partial^2 u_y}{\partial y^2} + G\frac{\partial^2 u_y}{\partial x^2} + (\lambda + 2G)\frac{\partial^2 u_x}{\partial x \partial y} + \alpha \frac{\partial P}{\partial y} = 0 \quad (16)$$

Creep Deformation

Long-term deformation is captured using a time-dependent creep formulation based on the Bailey-Norton power law:

$$\varepsilon_{cr} = \frac{3}{2} A n e^{-\frac{Q}{RT}} \sigma_e^{n-1} s_{ij} t^{m-1}, \quad (17)$$

where Q is activation energy, R is gas constant, T is temperature, s is deviatoric stress (the difference between principal stresses and hydrostatic stress) and σ_e is effective stresses. A, n and m are material constants.

The deviatoric stress is:

$$s_{ij} = \sigma'_{ij} - \frac{1}{2}(\sigma'_{xx} - \sigma'_{yy})\delta_{ij} \quad (18)$$

The total strain:

$$\varepsilon_{ij}^{total} = \varepsilon_{ij}^{elastic} + \varepsilon_{ij}^{creep} \quad (19)$$

The volumetric stain:

$$\varepsilon_v^{total} = \varepsilon_{xx}^{total} + \varepsilon_{yy}^{total} \quad (20)$$

Porosity is updated as follows:

$$\varphi_{new} = \varphi_o (1 + \varepsilon_v^{total}) \quad (21)$$

And the permeability is updated using Carman-Kozeny equation, presented in [equation 17](#):

$$\frac{k}{k_o} = \left(\frac{\varphi}{\varphi_o}\right)^n \cdot \left(\frac{1 - \varphi_o}{1 - \varphi}\right)^m \quad (22)$$

Results and Discussion

The numerical simulations were conducted using a two-dimensional coupled poromechanical model developed from first principles to assess the hydro-mechanical response of a deep saline aquifer under hydrogen injection and withdrawal operations. The model domain represents a square reservoir of 100 m × 100 m, discretized into 45 × 45 uniform cells with each cell has a lateral dimension of 2.22 m and a vertical thickness of 1.0 m. Initial porosity was uniformly assigned as 0.20, with a pore compressibility of 1 × 10⁻⁷ Pa⁻¹. Absolute permeability was spatially heterogeneous, generated from a uniform random distribution in the range of 1 to 700 mD, and converted to SI units (9.869233 × 10⁻¹⁶ m²). The rock framework was characterized by a Young's modulus of 1.0 × 10¹⁰ Pa, Poisson's ratio of 0.25, and Biot coefficient of 0.8. Corresponding Lamé parameters were computed as $\lambda = 6.67 \times 10^9$ Pa and $G = 4.0 \times 10^9$ Pa. Initial reservoir pressure was set to 3000 psi (20.684 MPa).

Laboratory Case Study

The stress-strain creep deformation power-law model was validated using a 5 × 5 × 5 homogeneous sandstone sample with a Young's modulus of 40 GPa and a Poisson's ratio of 0.17, calibrated against laboratory data reported by Yang and Hu (2018). In this study, only the elastic parameters (Young's modulus and Poisson's ratio) were varied to reflect different sandstone compositions, while all other parameters remained fixed (presented in [Table 1](#)).

Table 1—The input data used for the laboratory case study.

Input Data	
Activation Energy (Q), J/mol	20,920
Material Constant (n)	1.55
Material Constant (m)	1.005
Material Constant (A)	1.63×10^{-21}
Young's Modulus (E), GPa	40
Poisson Ratio (ν)	0.17
Gas Constant J/mol/K	8.314

The model was applied to a heterogeneous sandstone rock sample to evaluate creep deformation under elevated temperature and stress cycling. A three-dimensional grid system was used to simulate the evolution of creep strain across three loading cycles followed by two relaxation periods at 300 °C. Mineral composition was represented as a mixed distribution generated in MATLAB, with elastic parameters (Young's modulus and Poisson's ratio) calibrated to reflect the heterogeneous fabric of the rock. Figure 1 illustrates the spatial distribution of creep strain at the end of each cycle, highlighting both the cumulative deformation and the recovery during relaxation.

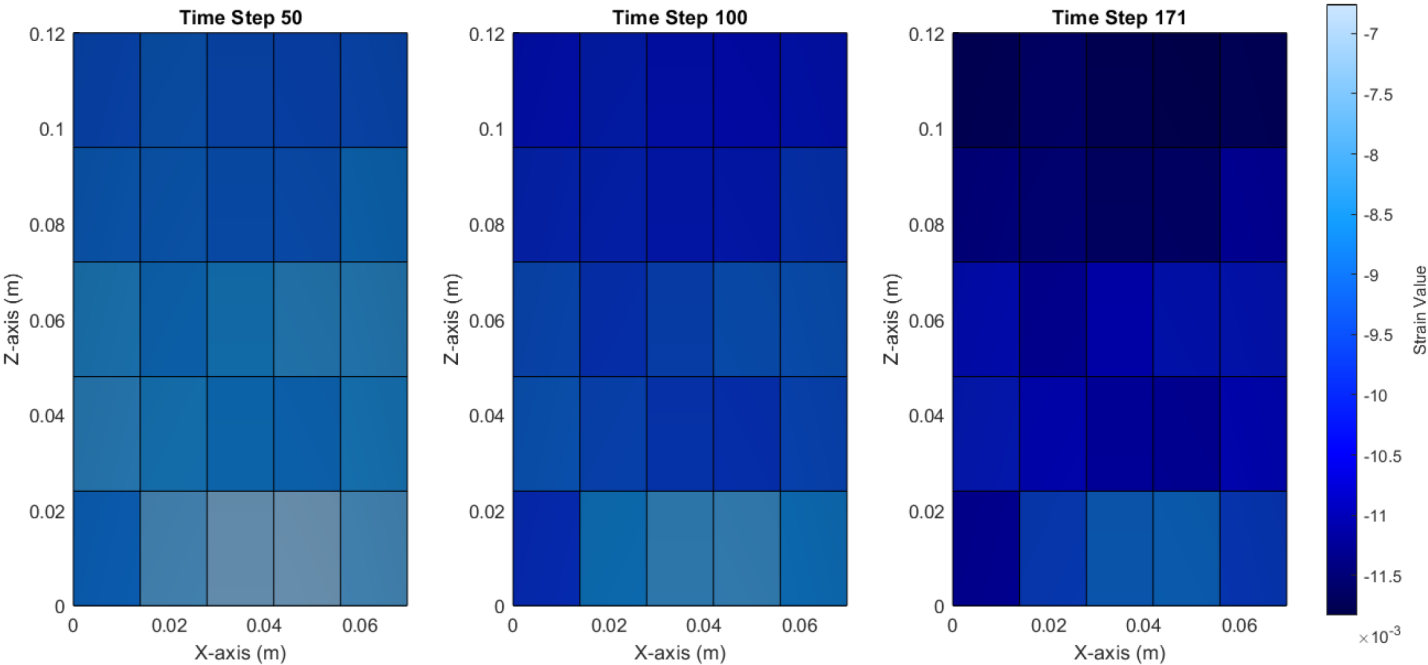


Figure 1—Creep Deformation strain for a the sandstone sample.

The heterogeneous sandstone sample exhibited a total creep strain of 5.6×10^{-3} after three cycles, with deformation concentrated in regions of lower stiffness. The variability in local elastic properties controlled the rate and magnitude of creep, as areas with lower Young's modulus deformed more readily under sustained stress. Poisson's ratio also influenced the distribution of strain, producing localized zones of non-uniform creep where stress concentrations were greatest.

Single Injection Case

Before examining multi-cycle responses, the model was first applied to a homogeneous formation to establish baseline hydro-mechanical behavior under a single injection event. In this configuration, material and hydraulic properties were spatially uniform, allowing the response to be attributed solely to the coupled poroelastic and flow processes rather than heterogeneity effects. The initial assessment focused on the pressure field evolution and the associated volumetric strain distribution, which together provide a first-order indication of the mechanical influence zone around the injector. These results are presented in Figure 2.

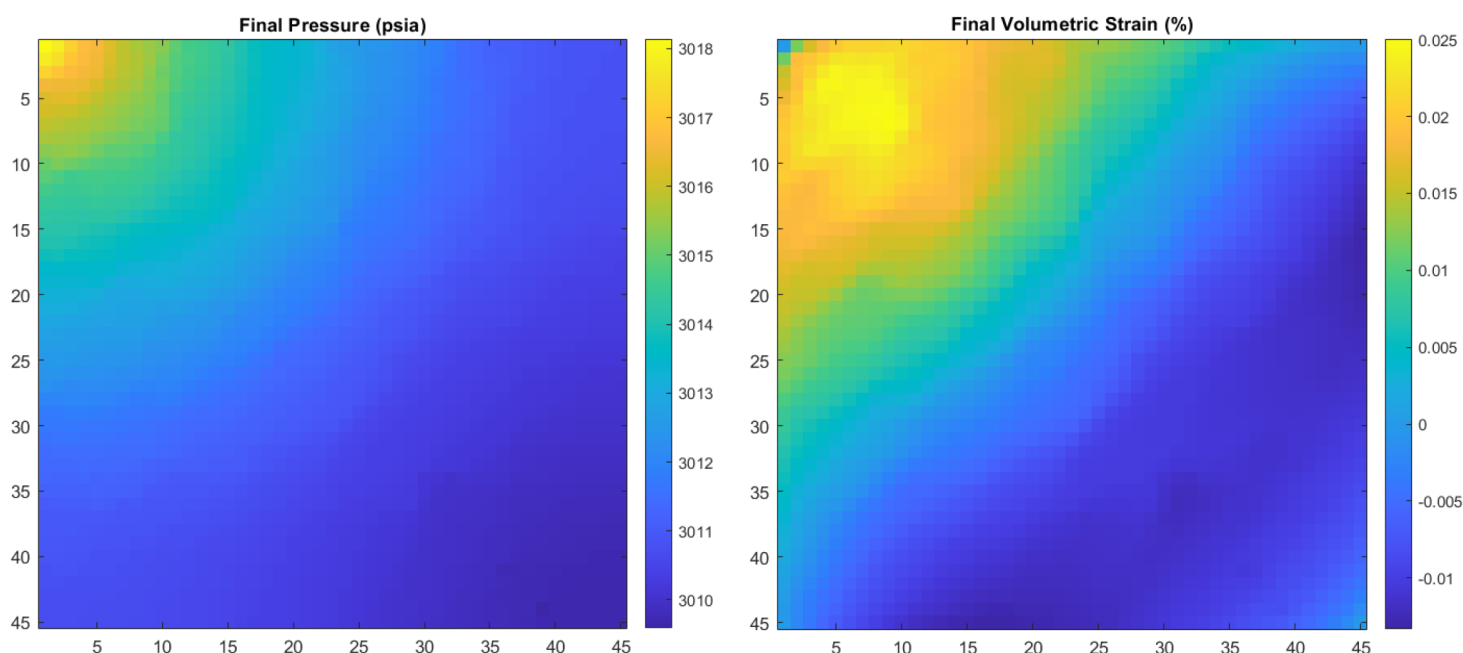


Figure 2—The final pressure and volumetric strain distribution of the homogenous case study.

The simulated pore pressure distribution during the 30-day hydrogen injection period displays a distinct radial gradient centered on the injection cell. A pronounced near-well pressure elevation arises from the limited mobility contrast between hydrogen and the resident brine, combined with the localized nature of the injection, which generates a high-pressure cone that diminishes progressively with distance from the source. The pressure front advances smoothly across the reservoir, with gradients becoming shallower toward the far field where pore pressures remain close to the initial conditions. This gradual attenuation reflects the combined influence of finite injection rates and the heterogeneous permeability structure, both of which restrict the spatial extent of hydraulic perturbations within the injection timeframe. The volumetric strain field exhibits a similar spatial footprint, with deformation concentrated around the injection cell. The strong correlation between pore pressure elevation and volumetric strain confirms that stress redistribution is driven primarily by effective stress reduction, leading to localized elastic expansion of the pore network. Comparable behavior was observed when modeling fluid flow independently of the geomechanical response, where the propagation of the pressure front and its attenuation with distance followed the same radial pattern.

From a geomechanical standpoint, the elastic volumetric strain is consistent with Biot's poroelastic framework: elevated pore pressure reduces mean effective stress, leading to matrix dilation where the stiffness-pressure coupling is strongest. The absence of significant tensile strain implies that the applied injection rate remains within safe operational limits for the given mechanical properties. The effective stresses and their distribution across the reservoir modeled section are presented in Figure 3.

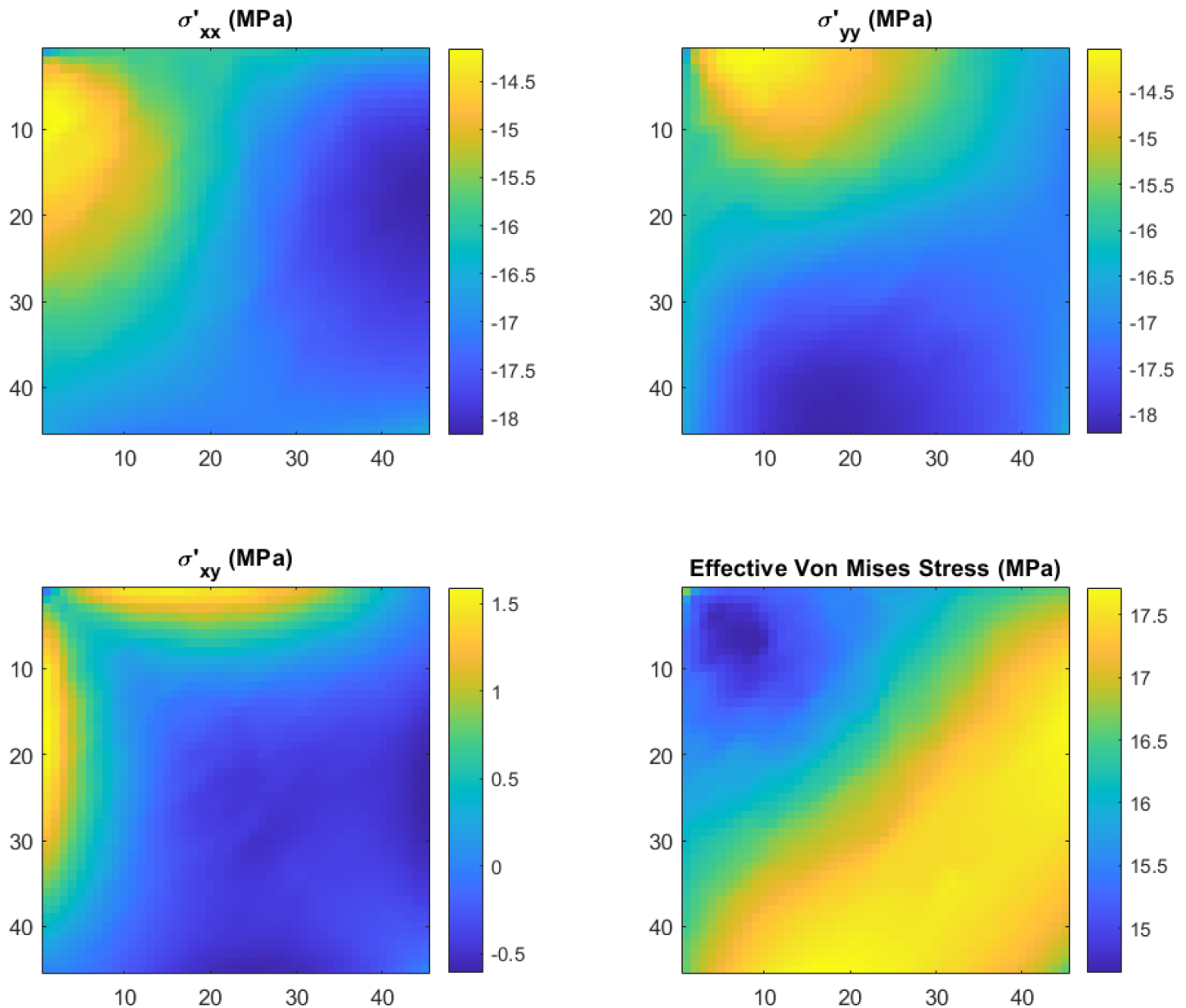


Figure 3—Effective stress components at the final timestep after hydrogen injection in an aquifer.

The effective stress distributions (Figure 3) demonstrate the redistribution of mechanical load within the formation in response to pore pressure elevation. Both the σ'_{xx} and σ'_{yy} components exhibit reductions of up to 14.5 MPa in the vicinity of the injection cell compared to the initial state, consistent with the direct offset of total stress by pore pressure. This reduction is most evident along the radial direction, in line with the isotropic nature of injection-driven pressurization. The effective von Mises stress field further reveals an annular zone of elevated shear stress encircling the pressure maximum. This pattern underscores the dual effect of injection: while pore pressure diminishes the mean effective stress, it concurrently amplifies shear stress concentrations in the transitional zone between high and low pressure. Such regions are critical for long-term stability evaluation, as repeated injection-withdrawal cycles may progressively weaken the rock framework. A comparable response is evident in the deviatoric stress distributions shown in Figure 4, which similarly highlight localized shear intensification around the injection zone.

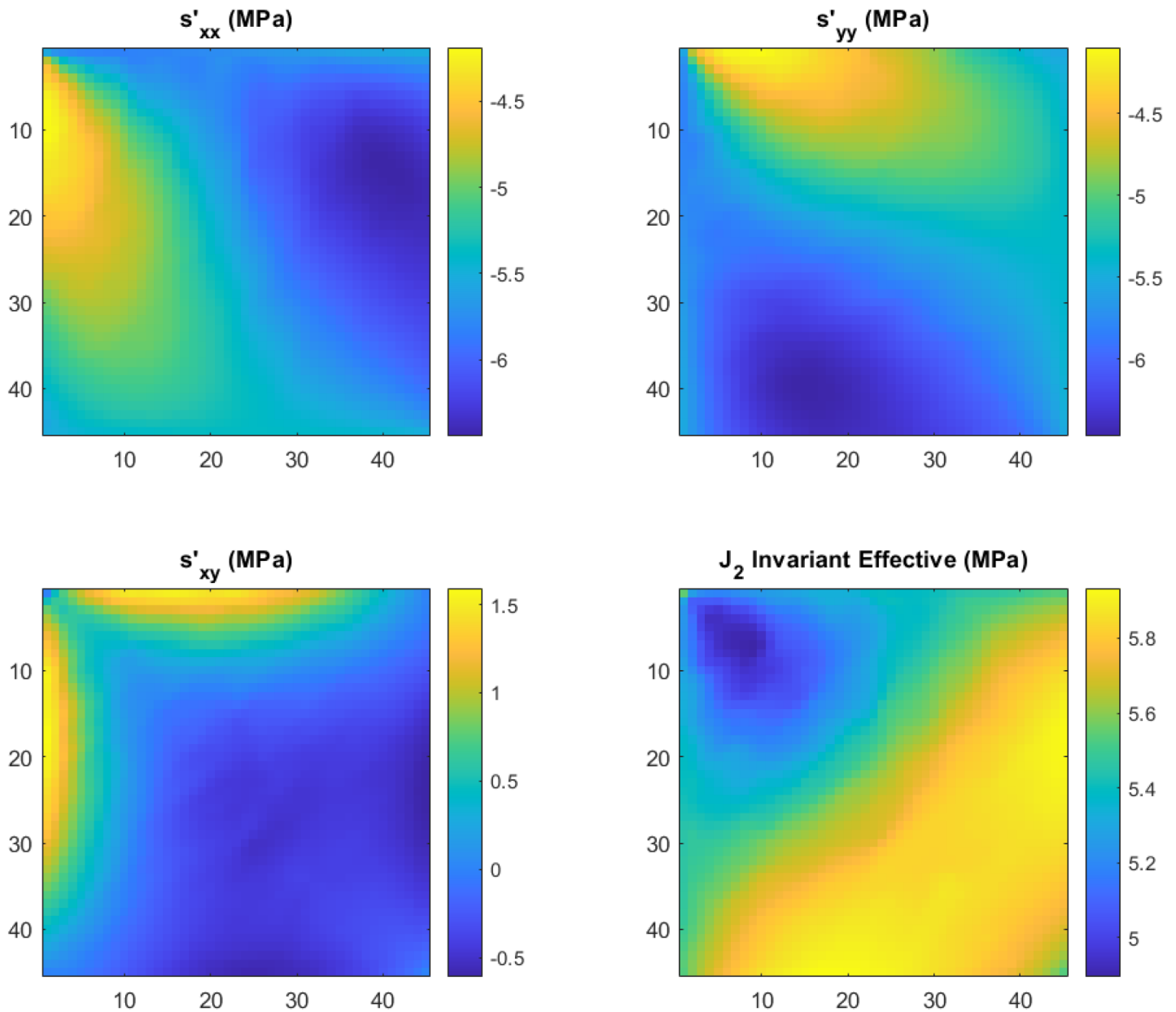


Figure 4—Effective deviatoric stress components at the final timestep after hydrogen injection in an aquifer.

The total deviatoric stress fields (Figure 4) provide additional clarity on the shear regime activated during injection. The s_{xx} and s_{yy} components form symmetric lobes of opposite sign around the injection cell, with peak absolute magnitudes of approximately 0.12 MPa. The shear component s_{xy} exhibits a quadrant-like pattern of alternating polarity, reaching values near 0.05 MPa, which confirms that deformation extends beyond volumetric strain and includes directional distortions of the rock matrix. This interpretation is reinforced by the J_2 invariant distribution, which reveals an annular zone of maximum shear intensity aligned with the effective von Mises stress peak. Such patterns are typical of radial pressurization in an elastic-poroelastic medium, where the greatest deviatoric stresses emerge not at the injection point itself but in the surrounding transition zone, where stress gradients and differential displacements are most pronounced.

A heterogeneous case was introduced to assess lateral hydrogen migration within the simulated reservoir. Permeability values were varied across the domain in the range of 1-700 mD, with the corresponding porosity distribution recalculated using the Carman-Kozeny relationship. Figure 5 presents the spatial permeability pattern together with the simulated hydrogen flow over the 30-day injection period.

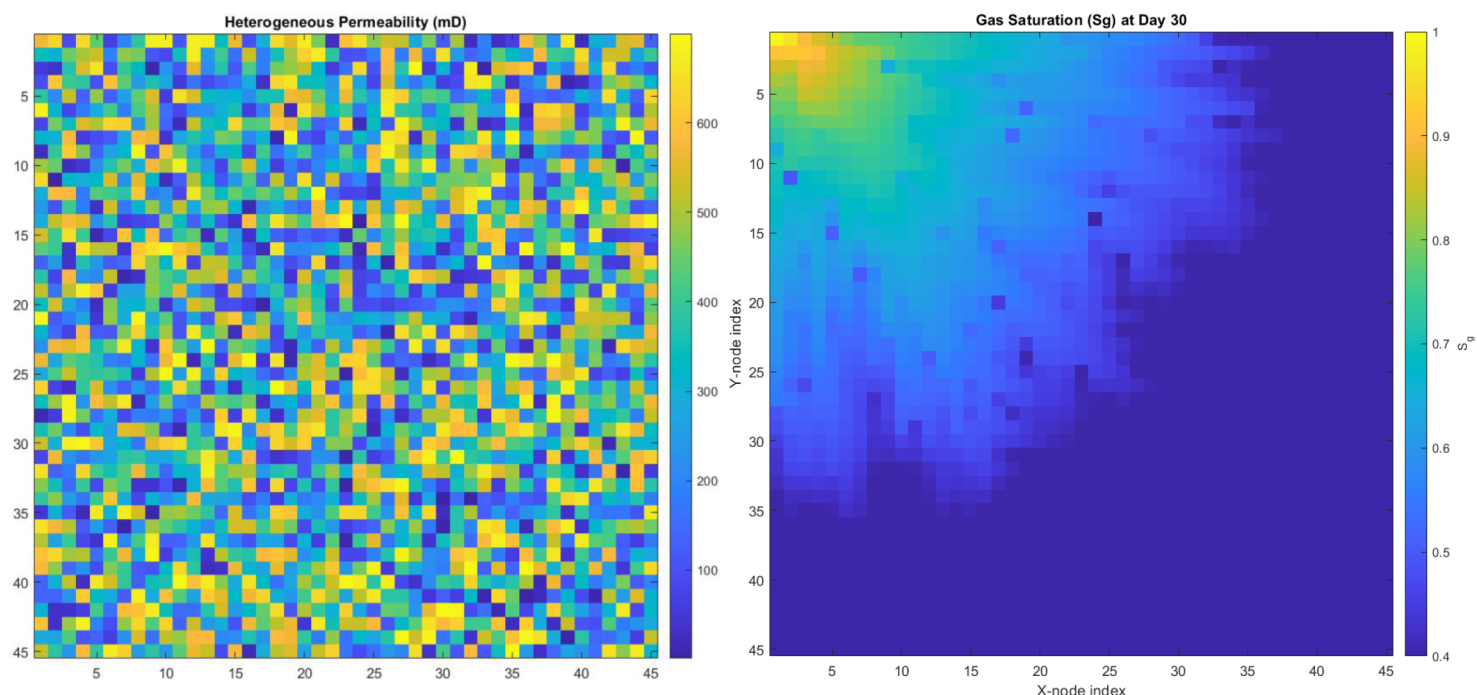


Figure 5—The heterogeneous permeability map and the fluid distribution map at day 30.

Following the evaluation of the pressure, volumetric strain, and stress responses, the next stage of the analysis focuses on the evolution of porosity and permeability within the reservoir during the single injection cycle. These hydraulic properties are directly influenced by deformation-induced changes in pore structure and are critical for understanding injectivity trends and long-term formation behavior. Figure 6 examines the evolution using three representative permeability locations within the model domain: the injection cell, a mid-domain cell, and a far-field cell and the porosity across these similar cells.

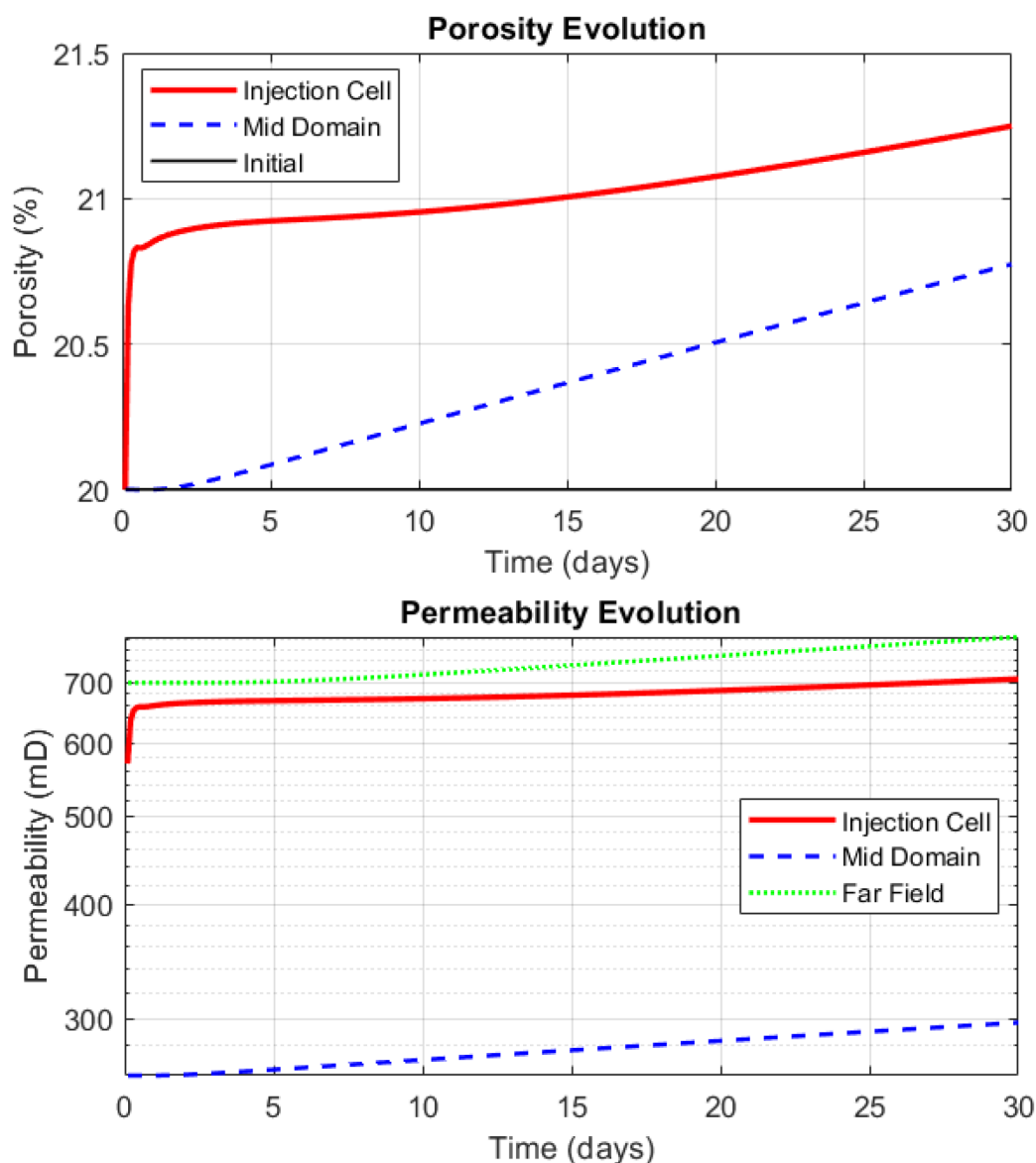


Figure 6—The evolution using three representative permeability locations within the model domain: the injection cell, a mid-domain cell, and a far-field cell and the porosity across these similar cells.

The single-cycle results illustrate the interplay between pore pressure, elastic deformation, and property evolution in a coupled hydro-mechanical system. Elevated pore pressure reduces effective stress, inducing elastic dilation that, in turn, enhances permeability near the well. However, the presence of a shear stress annulus in both the effective and deviatoric stress fields highlights regions that could be mechanically sensitive under repeated cycling.

Two-Cycles Injection Scheme

Extending the simulation to two full injection-production cycles highlighted the cumulative influence of cyclic loading on reservoir mechanical and hydraulic behavior. During the second injection phase, pressure accumulation near the injector exceeded that of the first cycle, a result of residual deformation carried over from the initial loading. The volumetric strain distribution reflected this memory effect, where irreversible creep-driven compaction led to a gradual reduction in mechanical stiffness in the near-well region. When creep deformation was explicitly incorporated into the model, the progressive nature of volumetric strain became more evident. Sustained stress perturbations, particularly adjacent to the injection point where

effective stress variations are greatest, promoted time-dependent compaction. This creep response operated alongside elastic deformation, generating a ratcheting effect whereby total strain increased with each successive cycle despite identical pressure perturbations. Figure 7 illustrates this phenomenon, comparing the cumulative contribution of creep to the elastic and total strain responses.

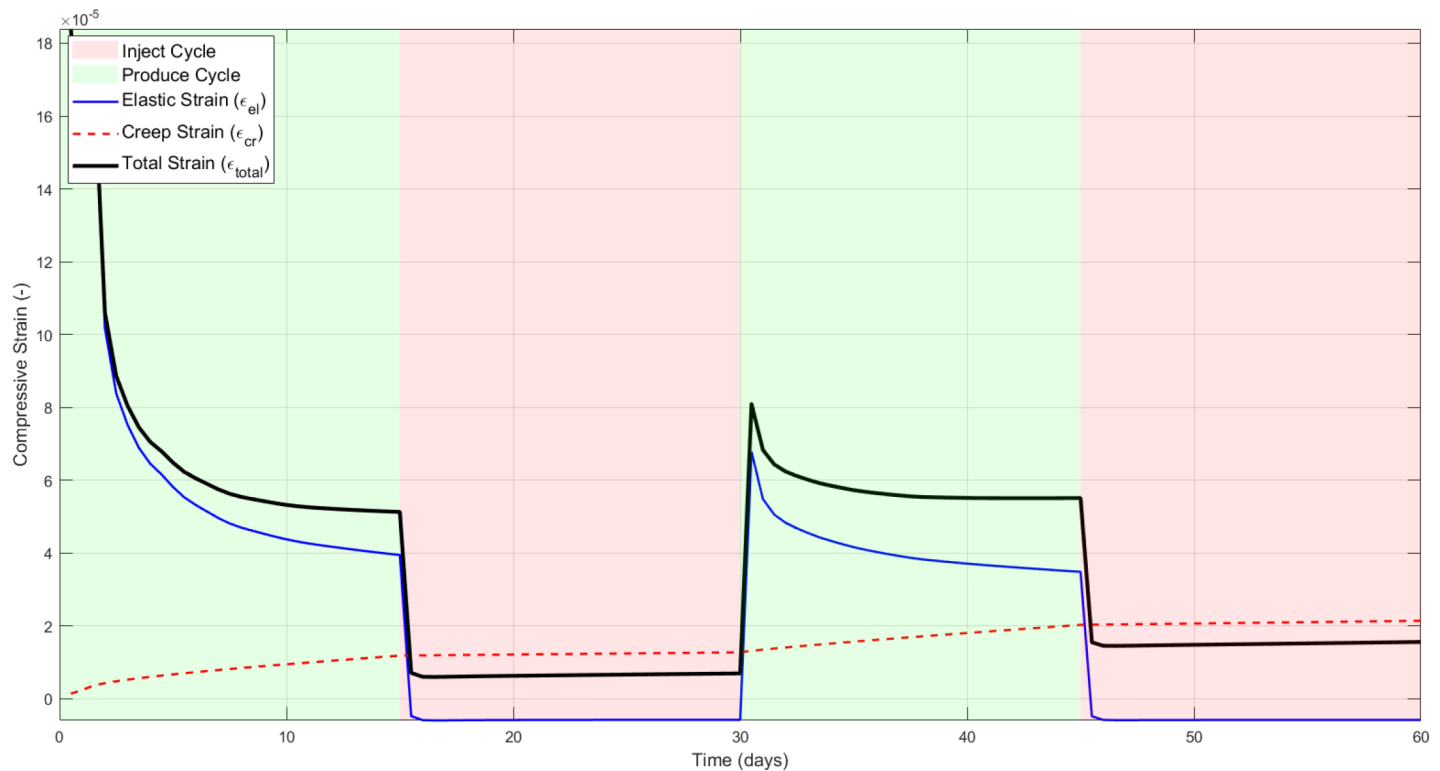


Figure 7—The effect of creep deformation presented by creep strain and in comparison, to elastic and total strain.

The presence of creep also slightly shifts the balance between recoverable and permanent deformation: elastic components dominate during the initial loading phase of each cycle, but the residual inelastic strain from creep accumulates in an irreversible manner. Over extended operational timelines, this could alter the long-term evolution of porosity and permeability, potentially increasing injectivity but also modifying stress paths in ways that may influence mechanical stability. This effect can affect crucial rock properties, mainly presented by porosity and permeability, as illustrated in Figure 8.

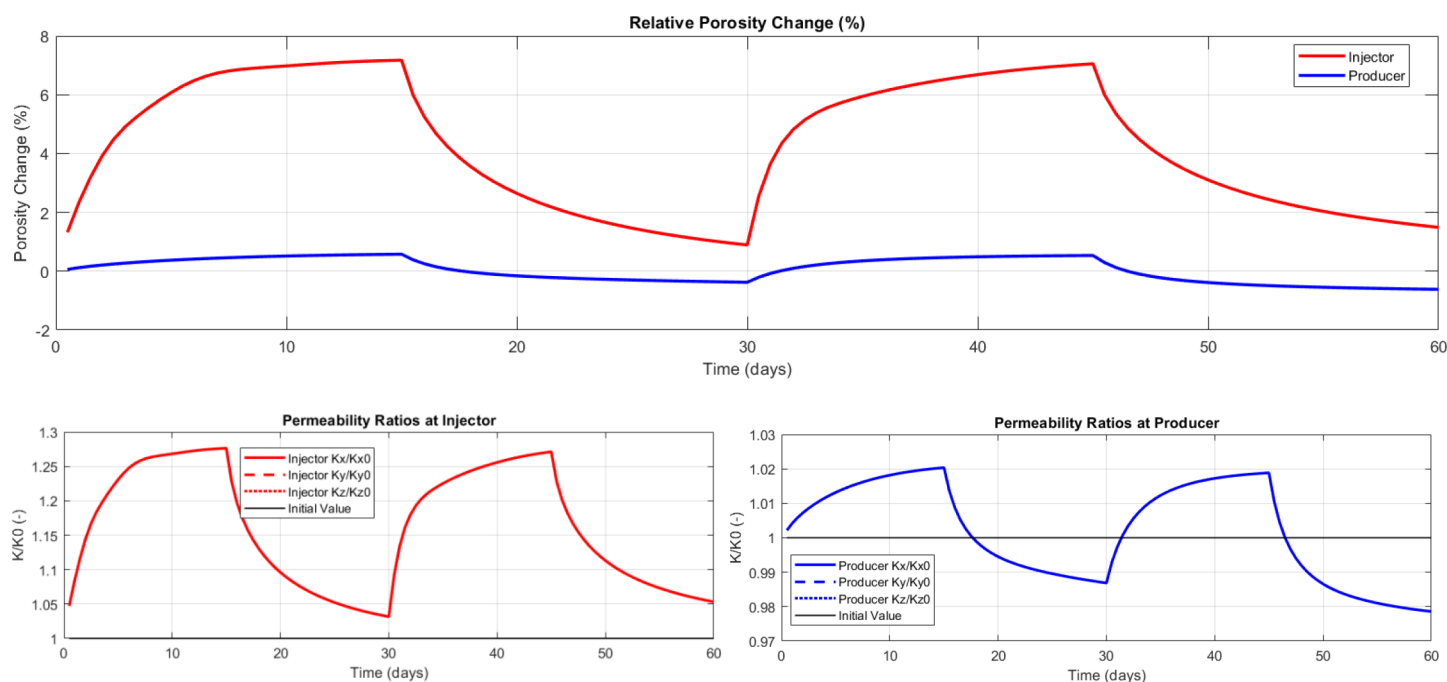


Figure 8—Permeability and porosity fluctuations during cyclic injection and withdrawal cycles.

The porosity distribution after two complete injection-production cycles demonstrates the cumulative effect of cyclic loading, with irreversible deformation concentrated near the injection cell. Whereas the first cycle generated a modest porosity increase confined to the near-well region, the second cycle broadened the affected zone both in magnitude and radial extent. At the injection cell, porosity increased by more than 6% relative to the initial state, more than doubling the enhancement observed after a single cycle. This sustained porosity growth following depressurization reflects the combined action of elastic dilation during injection and inelastic creep-driven compaction that is not fully reversible. The effect decreases with distance from the injector: mid-domain cells display small but measurable increases of around 1%, while far-field porosity remains largely unchanged. These localized yet permanent modifications of the pore network suggest improved injectivity during subsequent cycles, but also raise concerns for containment security if comparable processes occur in sealing formations. Permeability evolution exhibits similar cyclic sensitivity, with contrasting trends between injection and production wells. At the injector, directional permeability ratios rise sharply during pressurization as pore dilation reduces effective stress. The x-direction shows the largest increase, reaching a ratio of 1.27, equivalent to a 27% gain compared with the initial value. This anisotropy likely reflects alignment with the prevailing pressure gradient and displacement field around the wellbore. During withdrawal, permeability declines as effective stress rises, yet values never return fully to baseline, leaving behind a residual gain attributed to inelastic deformation and creep accumulation. The hysteretic behavior is clear: permeability responds more rapidly to injection than to compaction during production.

At the producer, permeability variations are more subdued. Ratios peak at about 1.021 (2.1% increase) during injection before falling slightly below unity during withdrawal. This limited response reflects the lower stress and strain perturbations relative to the injection zone. Nevertheless, the cyclic trend remains apparent, demonstrating that even low-strain regions exhibit hydraulic sensitivity to pressure cycling. After two full cycles, permeability ratios at the producer stabilize slightly below their initial values (≈ 0.98), indicating mild compaction-induced transmissivity loss. Collectively, these results highlight a fundamental contrast: injection wells are more susceptible to cumulative permeability enhancement, whereas production wells tend toward gradual compaction and capacity reduction under repeated cycling.

Conclusions

This study developed and applied a fully coupled two-dimensional poromechanical model, incorporating both elastic and time-dependent creep deformation, to evaluate the stability of a deep saline aquifer under cyclic hydrogen injection and withdrawal. The model was complemented by laboratory-scale insights into creep deformation and flow behavior, providing a multi-faceted perspective on underground hydrogen storage performance. The main findings can be summarized as follows:

- Laboratory observations confirmed that creep deformation is strongly temperature dependent, with sandstone samples showing accelerated strain rates at elevated temperatures, and that mineralogical heterogeneity enhances fracture nucleation, expediting the transition from microcracking to failure.
- Flow modeling demonstrated that pore pressure perturbations remain strongly localized near the injector, producing radial gradients that dissipate rapidly toward the far field, with strain fields closely mirroring pressure distributions.
- Stress redistribution during injection resulted in reductions of effective stress near the wellbore and the formation of an annular zone of elevated deviatoric stresses, highlighting potential zones of shear mobilization.
- Cyclic loading analysis revealed that residual porosity and permeability changes persist after production phases, with permeability increasing by up to 27% near the injector while minor compaction dominates at the producer.
- Creep inclusion introduced a ratcheting mechanism, whereby volumetric strain accumulates across successive cycles even under constant loading magnitudes, indicating irreversible long-term compaction.
- Operational implications point to a dual effect: localized property enhancements may improve injectivity, but long-term stability and sealing capacity could be compromised if creep-driven deformation alters stress paths or weakens shear-sensitive zones.

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